

Rafay Mahmood

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EDUCATION

Carnegie Mellon University

M.S. in Mechanical Engineering

Thesis: Exoskeleton Control and State Estimation | Coursework: Robot Learning, Optimal Control & RL, Modern Control Theory, Deep Learning

The Pennsylvania State University

B.S. in Mechanical Engineering

August 2025 – Present

Pittsburgh, PA

August 2021 – May 2025

University Park, PA

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Exoskeleton Researcher

MetaMobility Lab - Carnegie Mellon University

September 2025 – Present

Pittsburgh, PA

- Compared model-based and ML biotorque estimators for lower-limb joint torque across subjects, speeds (0.6–2.0 m/s), and inclines ($\pm 10^\circ$) against marker-based ground truth
- Built and benchmarked an online model-based human torque estimator under real-time data constraints, achieving $R^2 = 0.89$ and 0.127 N·m/kg RMSE against marker-based ground truth with no tuned parameters
- Built a real-time IMU pipeline on an NVIDIA Jetson—four BNO055 sensors over I2C at 46 Hz through a TCA9548A mux and custom integration board—with a VQF-based AHRS and functional sensor-to-segment calibration, achieving 41 ms end-to-end latency
- Derived the Euler-Lagrange dynamics of a 4-DoF foot-rooted biped symbolically and implemented closed-form M/C/N terms in Python, running 178x faster than the numerical model at 74 μ s per step

Undergraduate KINARM Robot Researcher

Sensorimotor Neuroscience and Learning Lab - The Pennsylvania State University

August 2024 – May 2025

University Park, PA

- Developed a nonlinear neuromuscular model in Simulink combining vestibular, proprioceptive, and saccadic eye-movement closed-loop feedback systems
- Designed and implemented an inverted-pendulum feedback-control model to simulate neuromuscular responses to mechanical perturbations and generate synthetic experimental data for the lab to compare with KINARM data

eVinci Component Design Intern

Westinghouse Electric Corporation

May 2024 – August 2024

Cranberry, PA

- Designed an encapsulation segmentation and an updated heat puck model using SolidWorks and Autodesk Inventor Pro and updated the instrumentation and shielding design document 2 months ahead of internal design review

PROJECTS

Golf Shot Optimization | Python, MuJoCo, CMA-ES

Feb 2026 – May 2026

- Built a MuJoCo golf simulation with aerodynamic drag, wind, and terrain-dependent rolling friction, and a CMA-ES framework to optimize launch parameters across three courses of increasing difficulty
- Benchmarked CMA-ES against differential evolution on identical courses, converging in 68% fewer simulation rollouts and 4x less wall-clock time

Controls Education with Maglev | C++, PID Control, Simulink, SolidWorks

October 2025 – December 2025

- Led programming of a PID controller for IR-sensed maglev and tuned gains for stable position control
- Modeled nonlinear system dynamics of levitating magnets in Simulink and iterated housing designs in SolidWorks

Student Rocket | C++, Teensy 4.1, Python, KiCad, SolidWorks

September 2023 – December 2023

- Developed embedded flight software in C++ on a Teensy 4.1 for real-time temperature and radiation sensing, with a custom KiCad avionics board

TECHNICAL SKILLS

Programming: Python, MATLAB, C++

Robotics & ML: Simulink, NumPy, pandas, PyTorch, MuJoCo, OpenSim

Embedded: Linux, NVIDIA Jetson, Teensy 4.1, I2C/CAN

Controls: PID, LQR, MPC, Nonlinear Control, State Estimation (VQF, Madgwick, NDO), Path Planning (A*), Trajectory Optimization, Derivative-Free Optimization (CMA-ES)

Software: SolidWorks, Autodesk Inventor, KiCad, Git, Vicon Motion Capture